TEST REPORT

Product Name	:	Smart Socket
Model Number	:	WI-02, WI-06

Prepared for	:	Ningbo Chuangguan Electric Appliance Co., Ltd
Address	:	Xinxin Village Industrial Park, LangXia Street, YuYao City, NingBo, ZheJiang, China
Prepared by	:	EMTEK (NINGBO) CO., LTD.
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Report Number	:	ENB2203180158W00303R
Date(s) of Tests	:	March 18, 2022 to March 28, 2022
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1. TEST RESULT CERTIFICATION

Applicant	:	Ningbo Chuangguan Electric Appliance Co., Ltd
Address	:	Xinxin Village Industrial Park, LangXia Street, YuYao City, NingBo, ZheJiang, China
Manufacturer	:	Ningbo Chuangguan Electric Appliance Co., Ltd
Address	:	Xinxin Village Industrial Park, LangXia Street, YuYao City, NingBo, ZheJiang, China
EUT	:	Smart Socket
Model Name	:	WI-02, WI-06
Trademark	:	N/A

Measurement Procedure Used:

APPLICABLE STANDARDS	
STANDARD	TEST RESULT
EN IEC 62311:2020	PASS

The device described above is tested by EMTEK (NINGBO) CO., LTD. to determine the maximum emission levels emanating from the device and the severe levels of the device can endure and its performance criterion. This report shows the EUT to be technically compliant with the EN IEC 62311:2020 requirements. The test results are contained in this report and EMTEK (NINGBO) CO., LTD. is assumed full responsibility for the accuracy and completeness of these tests.

This report applies to above tested sample only and shall not be reproduced in part without written approval of EMTEK (NINGBO) CO., LTD.

Date of Test : March 18, 2022 to March 28, 2022

Prepared by : June Gao /Editor

Reviewer : Vinay /Supervisor

Approve & Authorized Signer : Tony Wei/Manager

2. EUT DESCRIPTION

Product:	Smart Socket
Model Number:	WI-02, WI-06
Sample Number:	1#
WIFI	
WLAN Supported:	☒802.11b ☒802.11g ☒802.11n(20MHz channel bandwidth) ☐802.11n(40MHz channel bandwidth)
Modulation:	☒DSSS with DBPSK/DQPSK/CCK for 802.11b ☒OFDM with BPSK/QPSK/16QAM/64QAM for 802.11g/n
Frequency Range:	☒2412-2472MHz for 802.11b/g/n(HT20) ☐2422-2462MHz for 802.11n(HT40)
Number of Channels:	☒13 Channels for 802.11b/g/n(HT20) ☐9 Channels for 802.11n(HT40)
Max Transmit Power:	15.73 dBm
Antenna:	PCB Antenna
Antenna Gain:	-1.0 dBi,
Test Voltage:	AC230V, 50Hz
Date of Received:	March 18, 2022
Temperature Range:	-40°C ~ +85°C

Note: for more details, please refer to the User's manual of the EUT.

Modified Information

Version	Report No.	Revision Date	Summary
/	ENB2203180158W00303R	/	Original Report

3. FACILITIES AND ACCREDITATIONS

3.1 FACILITIES

All measurement facilities used to collect the measurement data are located at

EMTEK (SHENZHEN) C0., LTD.

Building 69, Majialong Industry Zone District, Nanshan District, Shenzhen, China

EMTEK (NINGBO) CO., LTD.

1F Building 4, 1177#, Lingyun Road, National Hi-Tech Zone, Ningbo, Zhejiang, China

The sites are constructed in conformance with the requirements of ANSI C63.7, ANSI C63.10 and CISPR Publication 32.

3.2 EQUIPMENT

Radiated emissions are measured with one or more of the following types of linearly polarized antennas: tuned dipole, biconical, log periodic, bi-log, and/or ridged waveguide, horn. Spectrum analyzers with preselectors and quasi-peak detectors are used to perform radiated measurements.

Conducted emissions are measured with Line Impedance Stabilization Networks and EMI Test Receivers.

Calibrated wideband preamplifiers, coaxial cables, and coaxial attenuators are also used for making measurements.

All receiving equipment conforms to CISPR Publication 16-1, “Radio Interference Measuring Apparatus and Measurement Methods.”

3.3 LABORATORY ACCREDITATIONS AND LISTINGS

Site Description EMC Lab.	:	**Accredited by CNAS** The Certificate Registration Number is L6666. The Laboratory has been assessed and proved to be in compliance with CNAS-CL01:2018 (identical to ISO/IEC 17025:2017) **Accredited by FCC** Designation Number: CN1302 Test Firm Registration Number: 436491 **Accredited by A2LA** The certificate is valid until May 31, 2023 The Certificate Number is 4321.03. **Accredited by Industry Canada** The Certificate Registration Number is CN0114 Company Number: 9469A
Name of Firm	:	EMTEK (NINGBO) CO., LTD..
Site Location	:	1F Building 4, 1177#, Lingyun Road, National Hi-Tech Zone, Ningbo, Zhejiang, China
Subcontractor information		
Name of Firm	:	EMTEK (SHENZHEN) CO., LTD.
Site Location	:	Building 69, Majialong Industry Zone, Nanshan District, Shenzhen, Guangdong, China

4. GENERAL PRODUCT INFORMATION

4.1 BASIC RESTRICTION

Reference Levels

Council Recommendation 99/519/EC Annex III
Reference levels for electric, magnetic and electromagnetic fields (0 Hz to 300GHz)

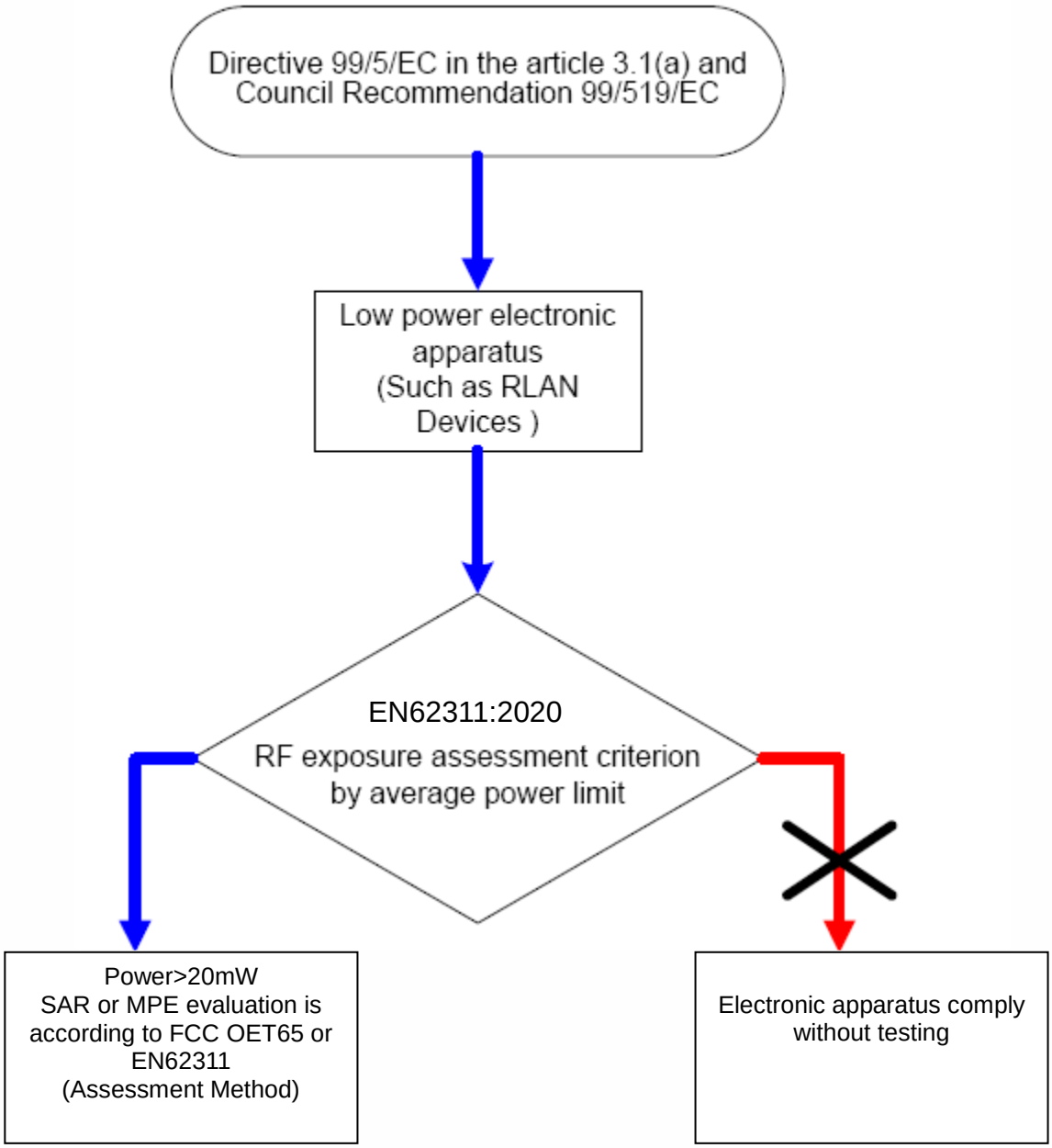
Frequency range	E-field strength (V/m)	H-field strength (A/m)	B-field (μT)	Equivalent plane wave power density Seq (W/m2)
0-1 Hz	-	3.2×10^{4}	4×10^{4}	-
1-8 Hz	10000	$3.2\times10^{4}/f^{2}$	$4\times10^{4}/f^{2}$	-
8-25 Hz	10000	4000/f	5000/f	-
0.025-0.8 kHz	250/f	4/f	5/f	-
0.8-3 kHz	250/f	5	6.25	-
3-150 kHz	87	5	6.25	-
0.15-1 MHz	87	0.73/f	0.92/f	-
1-10 MHz	$87/f^{1/2}$	0.73/f	0.92/f	-
10-400 MHz	28	0.073	0.095	2
400-2000 MHz	$1.375\ f^{1/2}$	$0.0037\ f^{1/2}$	$0.0046\ f^{1/2}$	f/200
2-300 GHz	61	0.16	0.2	10

Notes:

1. As indicated in the frequency range column.
2. For frequencies between 100kHz and 10 GHz, Seq, E2, H2 and B2 are to averaged over any six-minute period.
3. For frequencies exceeding 10 GHz, Seq, E2, H2, and B2 are averaged over any 68/1.05-minute period(in GHz).
4. No E-field value is provided for frequencies<1 Hz, which are effectively static electric fields. For most people the annoying perception of surface electric charges will not occur at field strengths less than 25 kV/m. Spark discharges causing stress or annoyance should be avoided.

4.2 EVALUATION ROUTINE

Low Power Electronic Apparatus for RF exposure evaluation routine



5. TEST RESULT

5.1 DETAILED RESULTS

5.1.1 Measurement of RF conducted Power

Band	EIRP Power
WiFi 2.4G	15.73 dBm

5.1.2 MPE Evaluation

$S = EIRP/ 4\pi R^2$

R = distance to the center of radiation of antenna (in meter) = 0.20 m

Note:
1) EIRP=P*G* Duty factor
2) P (Watts) = (10 ^ (dBm /10))/1000
3) G (Antenna gain in numeric) = 10^ (Antenna gain in dBi /10)
4) Duty factor
5) Π=3.142

Mode	Duty factor
WIFI	1

5.1.3 Summary of Results

The maximum power density at a distance of 0.2 m for WIFI is shown as below:

Mode	EIRP Power (dBm)	EIRP Power (W)	Duty factor	Calculated RF Exposure (W/ m²)	Limit (W/ m²)
WIFI 2.4GHz	15.73	0.037	1	0.0735	10

5.1.4 Measurement Uncertainty

Extended Uncertainty (k=2) 95% 0.5dB

*** End of Report ***

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